

WASP-135b

R-0107

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250223 · created 2026-07-16T03:39:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460730.015 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.046
Transit depth (Rp/Rs)^2	2.95 +/- 1.59 [%]
Orbital Inclination (inc)	78.43 +/- 2.96
Airmass coefficient 1 (a1)	1.084 +/- 0.02
Airmass coefficient 2 (a2)	-0.063 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#2 - [427, 442]
Optimal Aperture	3.41
Optimal Annulus	7.48
Transit Duration (day)	0.028 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.046 vs 0.139 (deviation 0.72σ)
Duration fit vs published	0.028 d vs 0.0687 d (deviation 1.57σ)
Mid-time O–C	20.6 min
Depth SNR	3.7
Residual scatter	1.82 %

Light curve

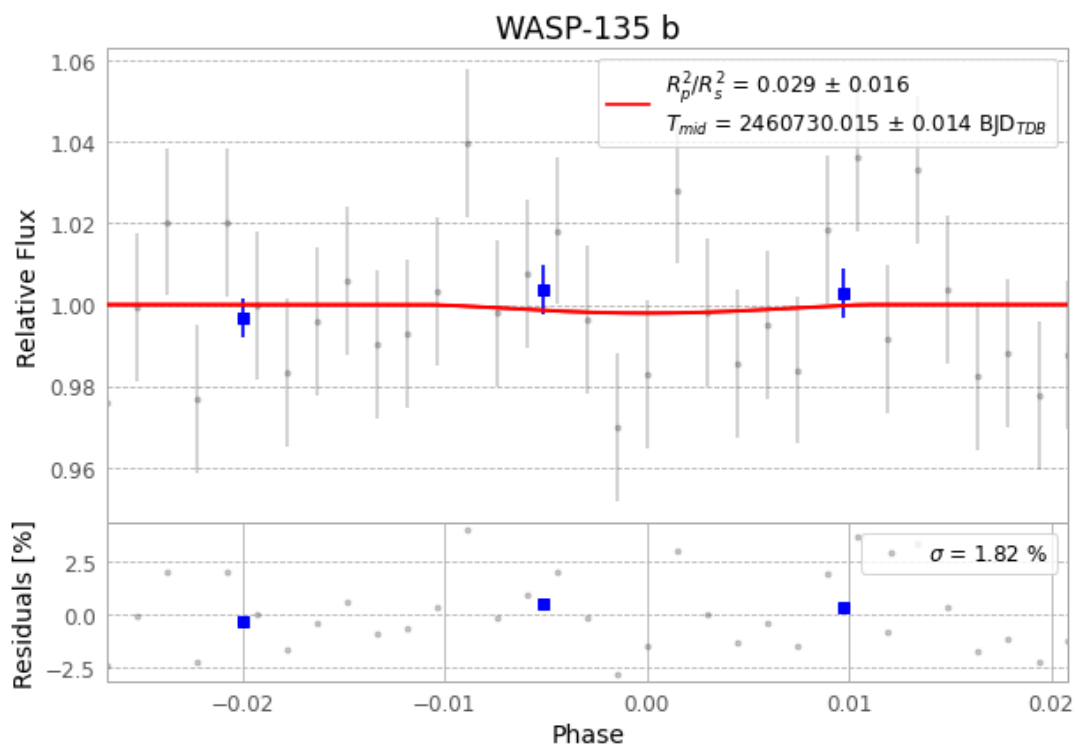


Figure 1 — FinalLightCurve_WASP-135 b_2025-02-23.png (SHA-256 5b1e54e94a216897...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1d8e2304e9d1b8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

33 FITS frames (22 MB), WASP-135250223112716.FITS → WASP-135250223130315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250223.log (13aea53646fe...), FinalLightCurve_WASP-135 b_2025-02-23.png (5b1e54e94a21...), FinalLightCurve_WASP-135 b_2025-02-23.csv (1bf71ffcf481...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:39Z.